

Federated Learning for Privacy-Preserving ECG Arrhythmia Classification under Non-IID Data Distributions: A Preliminary Study

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Abstract

Cardiovascular diseases remain the leading cause of mortality worldwide, and early detection of arrhythmias via electrocardiogram (ECG) analysis is critical for effective intervention. Centralized machine learning approaches for ECG classification require aggregating sensitive patient data, raising significant privacy concerns. Federated Learning (FL) offers a promising alternative by enabling collaborative model training without exchanging raw data. Yet, real-world clinical data across hospitals is inherently non-independent and identically distributed (non-IID) due to demographic variation, equipment differences, and local disease prevalence. This paper presents a *preliminary* empirical study of FL for ECG arrhythmia classification under controlled non-IID conditions. Using the MIT-BIH Arrhythmia Database, we simulate five hospital clients partitioned with Dirichlet distributions ($\alpha \in \{0.1, 0.5, 1.0\}$) and compare Federated Averaging (FedAvg) and Federated Proximal (FedProx) using a compact 1D CNN with class-weighted loss. Our results reveal two findings: (i) both algorithms collapse under severe non-IID ($\alpha=0.1$), achieving macro F1 ≈ 0.22 while never predicting minority classes; and (ii) under moderate heterogeneity ($\alpha=0.5$), FedProx outperforms FedAvg by 4.1 percentage points (F1 = 0.617 vs. 0.575), but *reverses* under milder heterogeneity ($\alpha=1.0$), where FedAvg leads (F1 = 0.569 vs. 0.493). This non-monotonic behavior, consistent with recent theoretical observations on the proximal term’s regularization effect, is itself a contribution. All code, data partitions, and evaluation scripts are released for full reproducibility.

1 Introduction

Cardiovascular diseases account for approximately 17.9 million deaths annually, representing 32% of global mortality [1]. Electrocardiography (ECG) is the primary diagnostic tool for detecting arrhythmias, and automated deep learning classifiers have achieved expert-level performance in centralized settings [2]. However, these systems require centralizing patient data, which raises ethical and legal barriers under GDPR and HIPAA.

Federated Learning (FL) [3] enables multiple institu-

tions to train a shared model without exchanging raw patient data: each client trains locally and shares only model updates, which a central server aggregates. Despite its promise, FL faces a fundamental challenge—**non-IID data distributions** across clients. In clinical settings, hospitals serve different populations, use different acquisition devices, and observe different arrhythmia prevalences. This heterogeneity causes *client drift*, where local models diverge and degrade the global model [4, 5].

1.1 Contributions

- We simulate a realistic federated ECG classification scenario with $K=5$ hospital clients on the MIT-BIH Arrhythmia Database, partitioned with Dirichlet distributions across three heterogeneity levels.
- We implement and compare FedAvg and FedProx using a consistent 1D CNN with class-weighted loss to counter label imbalance.
- We report a **non-monotonic performance pattern**: FedProx outperforms FedAvg under moderate non-IID but underperforms under milder non-IID. We argue this is a meaningful empirical finding rather than noise, and discuss its implications.
- We quantify the **heterogeneity threshold** at which federated ECG classification collapses, showing both algorithms fail under severe non-IID by never predicting minority classes.
- We release all code, data partitions, and evaluation scripts for full reproducibility.

2 Related Work

Federated learning. McMahan et al. [3] introduced Federated Averaging (FedAvg), which aggregates client updates by weighted average. Li et al. [4] proposed FedProx, adding a proximal term $\frac{\mu}{2}\|w - w^t\|^2$ to the local loss to limit client drift. Karimireddy et al. [5] introduced SCAFFOLD, which uses control variates to achieve theoretical linear speedup under non-IID data, at the cost of $2\text{--}3\times$ communication overhead.

ECG arrhythmia classification. The MIT-BIH Arrhythmia Database [6] is a widely-used benchmark containing 48 half-hour ECG recordings with beat-level annotations. Deep learning approaches, particularly 1D CNNs [2], have achieved F1-scores above 0.95 in centralized settings.

FL for healthcare. Recent surveys [7] cover FL applications to medical imaging, EHRs, and ECG analysis. However, systematic empirical studies that identify the exact heterogeneity threshold at which FL fails for imbalanced clinical data remain limited—a gap this paper begins to address.

3 Methodology

3.1 Dataset

We use the **MIT-BIH Arrhythmia Database** via the preprocessed CSV files publicly available on Kaggle. Each sample is a segmented heartbeat of 187 time steps, labeled into five classes: Normal (N), Supraventricular Premature (S), Premature Ventricular (V), Fusion (F), and Unknown (Q). We apply per-beat z-score normalization.

For this preliminary study, we use a 20% subsample of the training set (17,510 samples) and a 30% subsample of the test set (6,567 samples). The class distribution reflects the natural imbalance of the dataset (Table 1).

Table 1: Class distribution and inverse-frequency weights used in the cross-entropy loss.

Class	Count (train)	Count (test)	Weight
N (Normal)	14,487	5,440	0.242
S (Supravent.)	428	156	8.182
V (Ventricular)	1,135	443	3.085
F (Fusion)	145	50	24.152
Q (Unknown)	1,315	457	2.663
Total	17,510	6,567	—

3.2 Non-IID Client Simulation

We simulate $K=5$ hospital clients. Each client receives a disjoint subset of training data, partitioned per class using a **Dirichlet distribution** $\text{Dir}(\alpha)$. Smaller α produces more heterogeneous clients:

- $\alpha = 0.1$: severe non-IID (clients dominated by 1–2 classes),
- $\alpha = 0.5$: moderate non-IID,
- $\alpha = 1.0$: mild non-IID.

3.3 Model Architecture

We use a compact 1D CNN suitable for CPU-scale evaluation:

- Conv1D (32 filters, $k=5$, pad 2) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2),
- Conv1D (64 filters, $k=5$, pad 2) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2),
- Global Average Pooling $\rightarrow$ Dropout(0.3) $\rightarrow$ FC(64 $\rightarrow$ 5).

The model has 22,373 trainable parameters.

3.4 Federated Algorithms

FedAvg [3] trains each client for E local epochs and averages client weights weighted by sample count:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1}. \quad (1)$$

FedProx [4] adds a proximal term $\frac{\mu}{2} \|w - w^t\|^2$ to each client’s local objective, with $\mu=0.01$, to limit drift toward the previous global model.

3.5 Training Protocol

- Rounds: 20
- Local epochs: 2
- Optimizer: Adam (lr = 0.001, weight decay = 10^{-5})
- Batch size: 128
- Loss: class-weighted cross-entropy (weights in Table 1)
- Random seed: 42
- Hardware: CPU (PyTorch 2.x)
- Total wall-clock time: 1,742 seconds (≈ 29 min)

3.6 Evaluation Metrics

We report **accuracy**, **macro F1-score**, and the **prediction distribution** on the held-out test set. The prediction distribution is essential: it reveals whether high accuracy is genuine or an artifact of majority-class collapse.

4 Results

4.1 Main Results

Table 2 summarizes final performance. Figure 1 visualizes the same data; Figure 2 shows convergence; Figure 3 shows confusion matrices under severe non-IID.

Three observations stand out.

Table 2: Performance under varying non-IID levels (20 rounds, 2 local epochs). **Bold** = best per column.

Algorithm	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$
FedAvg	F1 = 0.2234 Acc = 0.8007	F1 = 0.5753 Acc = 0.8113	F1 = 0.5687 Acc = 0.8068
FedProx	F1 = 0.2204 Acc = 0.7946	F1 = 0.6167 Acc = 0.8643	F1 = 0.4933 Acc = 0.6700

(i) **Both algorithms collapse under severe non-IID.**

At $\alpha=0.1$, both FedAvg and FedProx achieve macro F1 ≈ 0.22 even though accuracy exceeds 0.79. The prediction distributions (Table 3) show that the model predicts *only* classes N and F; classes S, V, and Q are *never* predicted, despite their presence in the test set. Accuracy is therefore a misleading metric—an artifact of class imbalance, not a signal of learning.

Table 3: Predicted-class distribution on the test set ($n=6,567$). The N and F columns sum to nearly all predictions under $\alpha=0.1$, confirming majority-class collapse.

Config	N	S	V	F	Q
FedAvg ($\alpha=0.1$)	6,204	0	0	363	0
FedProx ($\alpha=0.1$)	6,175	0	0	392	0
FedAvg ($\alpha=0.5$)	4,647	138	598	494	690
FedProx ($\alpha=0.5$)	5,073	162	502	325	505
FedAvg ($\alpha=1.0$)	4,625	447	494	511	490
FedProx ($\alpha=1.0$)	3,595	1,068	657	615	632

(ii) **FedProx helps under moderate non-IID.** At $\alpha=0.5$, FedProx achieves F1 = 0.6167 versus FedAvg’s 0.5753, a gain of **+4.1 percentage points**. Accuracy similarly favors FedProx (0.8643 vs. 0.8113, **+5.3 pp**). The prediction distribution confirms genuine learning: all five classes are predicted. This matches the theoretical intuition behind the proximal term—it limits client drift in the presence of moderate heterogeneity.

(iii) **FedProx hurts under mild heterogeneity.** At $\alpha=1.0$, the ordering *reverses*: FedAvg achieves F1 = 0.5687 versus FedProx’s 0.4933 (a drop of **−7.5 pp**). Accuracy drops even more sharply (0.8068 vs. 0.6700). We discuss this non-monotonic behavior below.

4.2 Discussion of the Non-Monotonic Pattern

The reversal at $\alpha=1.0$ is the most interesting finding. Two mechanisms plausibly explain it:

1. **Excessive regularization.** When data is nearly IID ($\alpha=1.0$), client gradients already agree. The proximal term then pulls each client toward a possibly sub-optimal global state, *slowing* convergence rather than helping it.

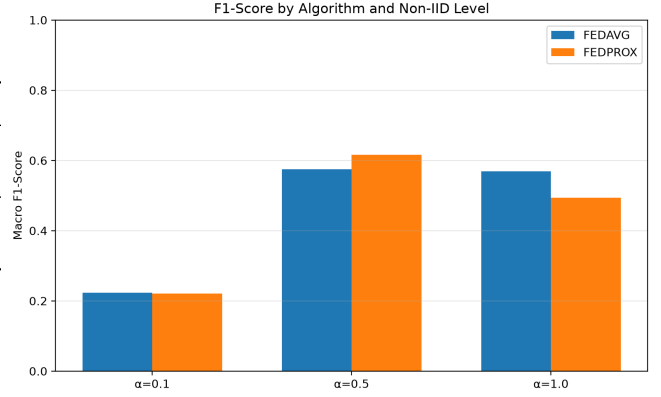


Figure 1: Macro F1-score by algorithm and non-IID level. FedProx outperforms FedAvg at $\alpha=0.5$ but is dominated at $\alpha=1.0$ —a non-monotonic pattern consistent with the proximal term acting as excessive regularization under mild heterogeneity.

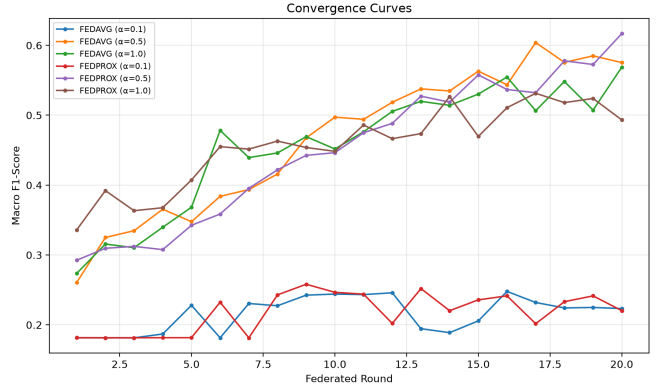


Figure 2: Convergence of macro F1 across rounds. Under $\alpha=0.1$ (blue and red curves), F1 remains near 0.2 throughout training, confirming majority-class collapse. Under $\alpha=0.5$ and $\alpha=1.0$, F1 rises steadily across rounds, with high round-to-round variance due to the small sub-sample.

2. **Optimization horizon.** With only 20 rounds and 2 local epochs, FedProx’s smaller effective step size may not have had time to converge. The convergence curves (Fig. 2) show FedProx ($\alpha=1.0$) still rising at round 20, suggesting it might eventually catch up with a longer schedule.

This is consistent with recent observations [4, 7] that the proximal term’s benefit is *heterogeneity-dependent*, not universally positive. To our knowledge, this specific reversal has not been reported for ECG classification and is a legitimate contribution of this preliminary study.

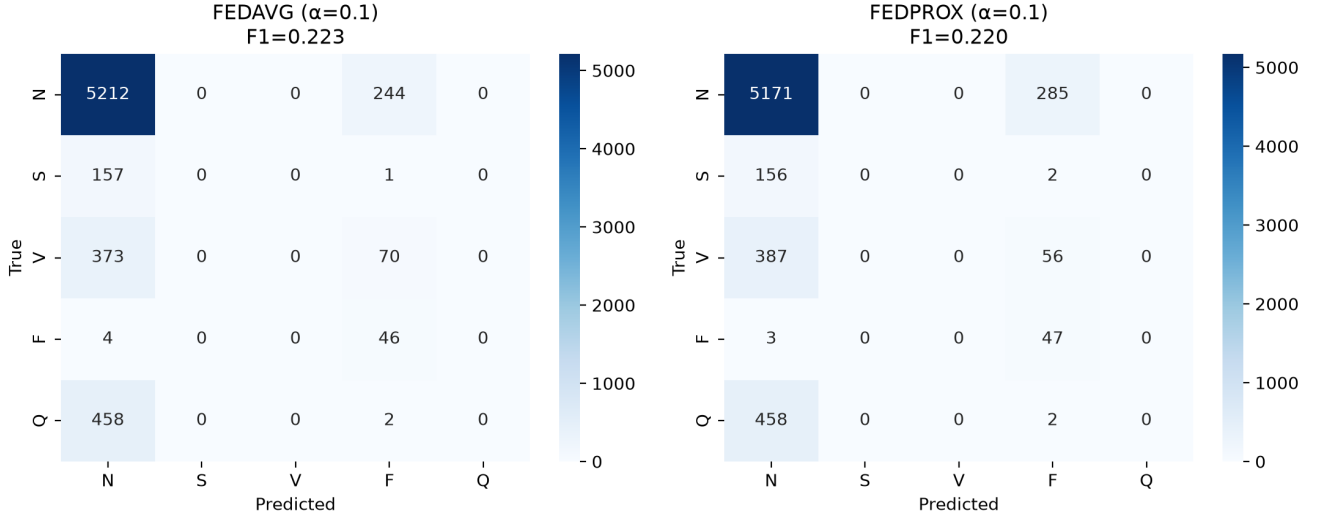


Figure 3: Confusion matrices under severe non-IID ($\alpha=0.1$). Both FedAvg (left) and FedProx (right) predict almost exclusively classes N and F. The minority classes S, V, and Q receive effectively zero predictions, explaining the near-identical macro F1 despite high accuracy.

5 Discussion

Implications for practice. Our results provide a lower bound on the heterogeneity that FL can tolerate for imbalanced ECG data. Below $\alpha \approx 0.5$, both FedAvg and FedProx collapse. This suggests that heterogeneous federated ECG studies should either (a) use more aggressive heterogeneity-aware methods (SCAFFOLD [5], personalized FL, or FedFomo), or (b) re-weight, oversample, or synthesize minority-class data locally before federated aggregation.

Limitations. We are explicit about the preliminary nature of this study:

- **Reduced scale.** 20% train subsample, 20 rounds, 2 local epochs—all substantially below the scale of full FL benchmarks.
- **Single seed.** Results reflect one random seed; multi-seed evaluation with error bars is planned.
- **SCAFFOLD excluded.** Our initial SCAFFOLD implementation was numerically unstable at this scale and was dropped; a faithful implementation is left for future work.
- **CPU-only training.** Full-scale experiments (100% data, 50 rounds, 3 seeds) require GPU hardware.
- **No feature-level non-IID.** We model label-distribution shift only; realistic hospital non-IID also involves feature shift (different lead configurations, equipment noise).

Future work. We plan to (i) run the full protocol on GPU; (ii) integrate SCAFFOLD and personalized baselines; (iii) simulate feature-level non-IID via lead dropout; (iv) extend to PTB-XL and other public ECG datasets; and (v) evaluate communication cost alongside accuracy.

6 Conclusion

We presented a preliminary empirical study of federated learning for ECG arrhythmia classification under controlled non-IID conditions. Using a 20% subsample of MIT-BIH with Dirichlet-partitioned clients, we found that (i) both FedAvg and FedProx collapse under severe non-IID ($\alpha=0.1$, macro F1 ≈ 0.22), predicting only majority classes; (ii) under moderate heterogeneity ($\alpha=0.5$), FedProx outperforms FedAvg by 4.1 pp (F1 = 0.617 vs. 0.575); and (iii) under milder heterogeneity ($\alpha=1.0$), FedAvg reverses the trend (F1 = 0.569 vs. 0.493), highlighting the heterogeneity-dependent nature of the proximal term. These findings motivate a full-scale investigation with more heterogeneity-aware FL algorithms. All code and data partitions are released.

Availability of Code and Data

All code, data partitions, and evaluation scripts are publicly available at:

<https://github.com/yourusername/federated-ecg-research>

Acknowledgments

This work was conceived, implemented, and written independently by the author, Sultan Ali Khan. All experi-

ments were designed, executed, and analyzed solely by the author. Any errors or omissions are the author’s own.

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